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Good Science

Drawing Valid Conclusions



Julian **Frattini**, Ph.D.

Postdoctoral Researcher

Chalmers University of Technology, Sweden

<https://julianfrattini.github.io/>



JulianFrattini/**bda4sci**

Public

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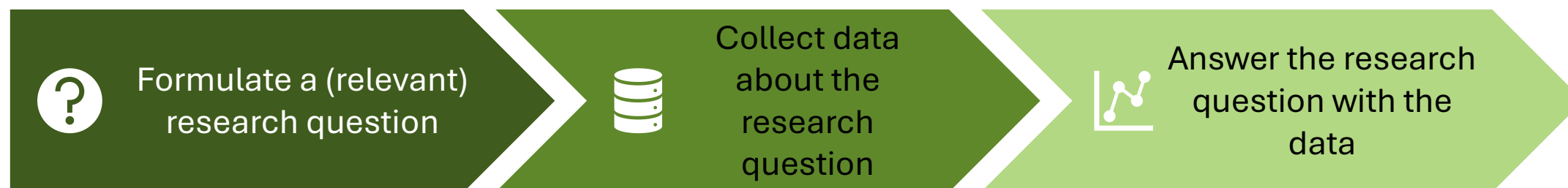


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Motivation

The Scientific Process

Considering the following, abstract scientific process:



Most of our concerns were related to formulating a research question and collecting data about it. But does that render the third step, its analysis, trivial?

Conclusion Validity

Even if we have ensured the internal, external, and construct validity of our study in the *data collection phase*, we might still draw the wrong conclusions from it.

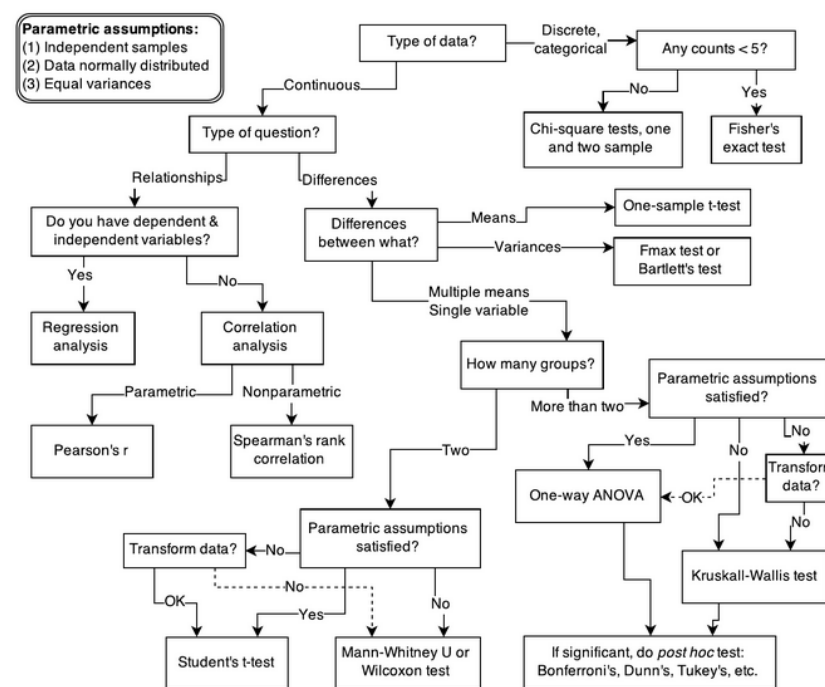


Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236, p. P3). Berlin: Springer.

Conclusion validity affects our “ability to draw the correct conclusion about relations between the treatment and the outcome of an experiment.”

Data Analysis Practices

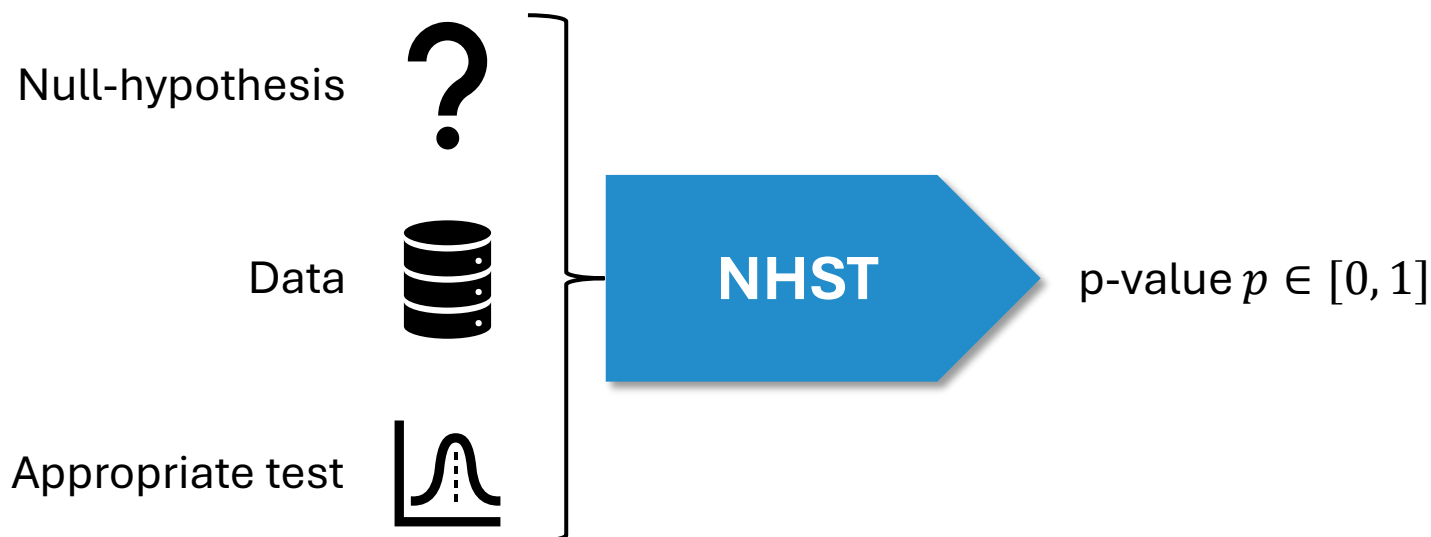
In many cases, data analysis boils down to the selection of the **most appropriate significance test**.



McElreath, Richard. *Statistical rethinking: A Bayesian course with examples in R and Stan*. Chapman and Hall/CRC, 2018.

Null-Hypothesis Significance Tests

Null-hypothesis significance tests (NHSTs) work roughly as follows:



Null-hypotheses are usually formulated such that they encode *no effect*, e.g., “ H_0 : The use of GenAI does not affect a developer’s productivity.”

Null-Hypothesis Significance Tests

To make a decision based on an NHST, the resulting p-value is compared with a significance level α :

- $p < \alpha$ represents a *statistically significant* result
- $p \geq \alpha$ represents *no* statistically significant result

Statistical significant results imply that two distributions are statistically unlikely to have come from the same population.

Null-Hypothesis Significance Tests

A p-value is a statement about the observed data.

” A p-value is the probability of observing data **at least as extreme as what you actually got**, *assuming the null hypothesis is true*.

A null-hypothesis (encoding “no effect”) receiving a p-value of 4% on observed data implies that – if that null-hypothesis were true (and there is indeed no effect) – there is a 4% chance we would observe the collected data.

Null-Hypothesis Significance Tests

A p-value is a **not** statement about the formulated hypothesis.

” A p-value is **not** the probability that the null hypothesis is correct, given what they observed.

However, this is usually the question of interest: how likely is there no/an effect of one variable on another? **NHSTs do not answer this question.**

Null-Hypothesis Significance Tests

Analysing data using only NHSTs has several issues:



Binary

NHSTs reduce complex data to binary, seemingly certain results



Normal

NHSTs distinguish only normal from non-normal data



Interpretability

The p-value is often misinterpreted



Arbitrary

There is nothing special about the significance level $\alpha = 5\%$



Think Bayesian

An Alternative Perspective on Probability

Probability

Next to the **frequentist** paradigm, the **Bayesian** paradigm of statistics assigns a different meaning to *probability*:

- In the **frequentist** paradigm, probability is a property of a repeatable random process. The probability of an event is the proportion of times it would occur if you repeated the experiment an infinite number of times under identical conditions. **Probability only assigns to random events, not hypotheses.**
- In the **Bayesian** paradigm, probability is a statement about *the state of knowledge of an observer*, not about physical processes in the world. A probability quantifies how confident a rational agent is that something is true, given their current information. **Probability assigns to belief, not to events** (as these are factual).

Bayes' Theorem

The Bayesian thinking is based on **Bayes' Theorem**:

	Likelihood	Prior
	of observing data d if the hypothesis h were true	probability that hypothesis h is true given no information
Posterior Probability that the hypothesis h is true given the observed data d	$\boxed{P(h d)} = \frac{\boxed{P(d h)} \cdot \boxed{P(h)}}{\boxed{P(d)}}$	
	Marginal likelihood i.e., the probability to observe the data d under all possible hypotheses	

Example

We are investigating a static code analysis tool. The tool has 90% sensitivity and 10% false positive rate. Only 5% of files actually have bugs. *What's the probability that a flagged file is buggy?*

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h : the file has a bug

d : the tool flags the file as buggy

Example

We are investigating a static code analysis tool. The tool has **90% sensitivity and 10% false positive rate**. Only 5% of files actually have bugs. *What's the probability that a flagged file is buggy?*

	Likelihood	Prior
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The marginal likelihood $P(d)$ can be calculated from the other data:

$$P(d) = P(d|h) \cdot P(h) + P(d|\neg h) \cdot P(\neg h)$$

$$0.9 \cdot 0.05 + 0.1 \cdot 0.95 = 0.14$$

Marginal likelihood

i.e., the probability to observe the data d under all possible hypotheses

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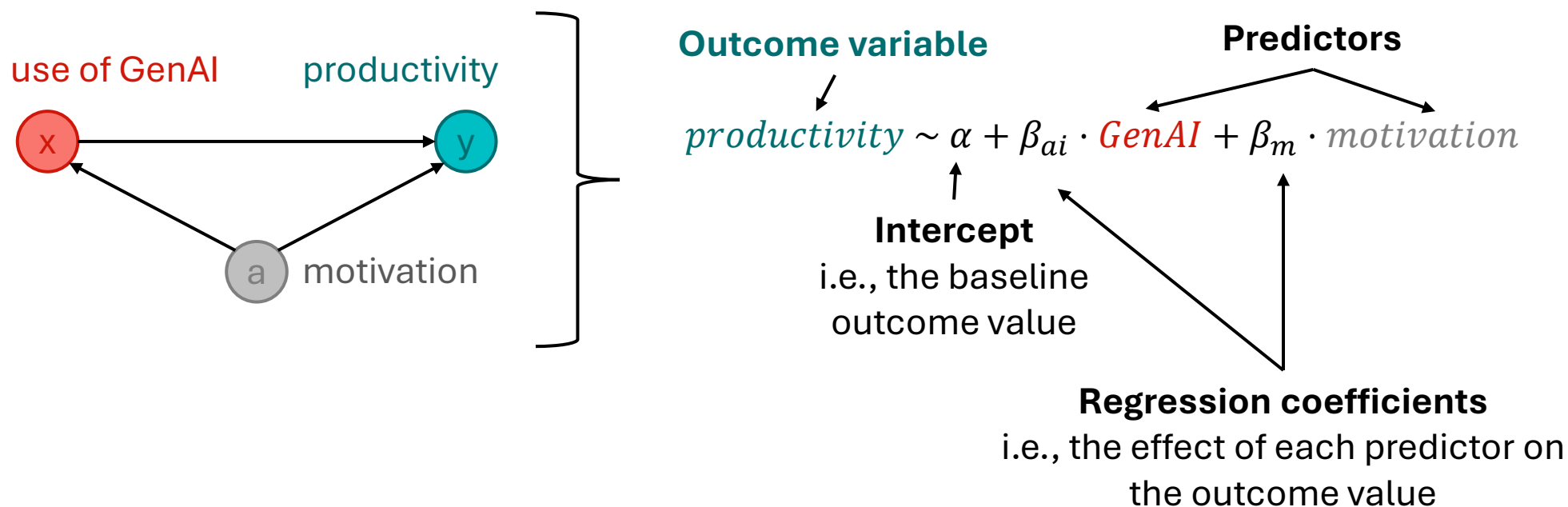
	Likelihood	Prior	
	of observing data d if the hypothesis h were true	probability that hypothesis h is true given no information	
Posterior Probability that the hypothesis h is true given the observed data d	$P(h d) = \frac{0.90 \cdot 0.05}{0.14} = 0.321$		
	Marginal likelihood i.e., the probability to observe the data d under all possible hypotheses		



Bayesian Data Analysis

Bayesian Regression Models

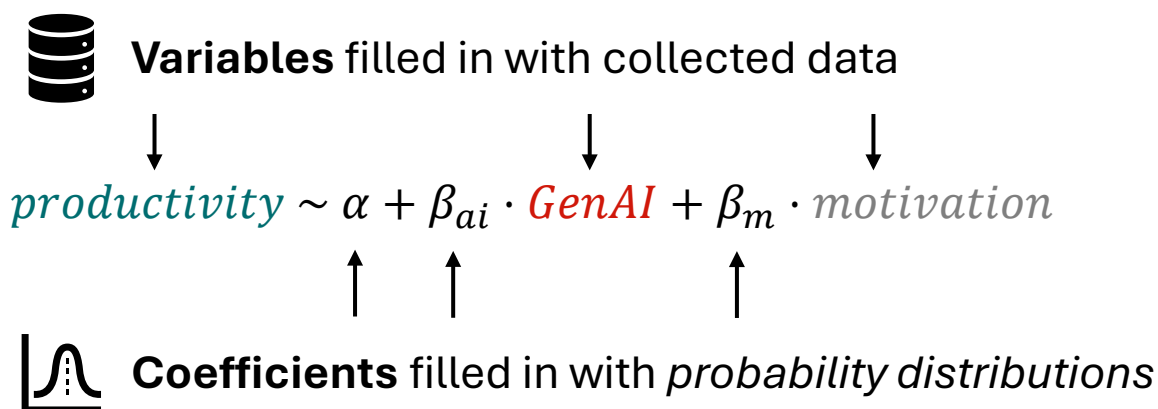
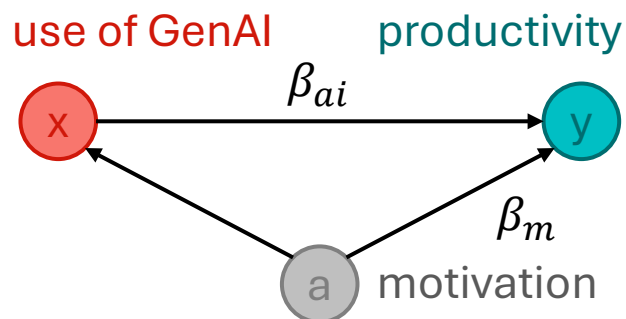
Utilizing Bayes' theorem to answer research questions requires to formulate **regression models** of the following (simplified) form:



Bayesian Regression Models

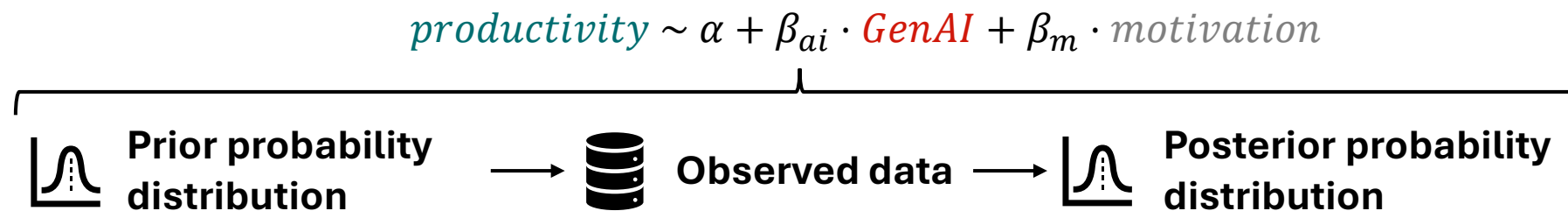
A regression model contains two types of terms:

1. **Variables** representing the values of events, and
2. **Coefficients** representing the impact of these variables on each other.



Bayesian Regression Models

The process of training a Bayesian regression model is based on applying Bayes theorem: in order for the **regression coefficients** to represent the impact of variables, **prior probability distributions** are updated in light of **the data** to yield **posterior probability distributions**.



Bayesian Regression Models

Selecting priors for a regression coefficient means assigning a probability distribution to it.

$$\textit{productivity} \sim \alpha + \beta_{ai} \cdot \textit{GenAI} + \beta_m \cdot \textit{motivation}$$

$$\alpha \sim N(\mu = 5, \sigma = 2)$$

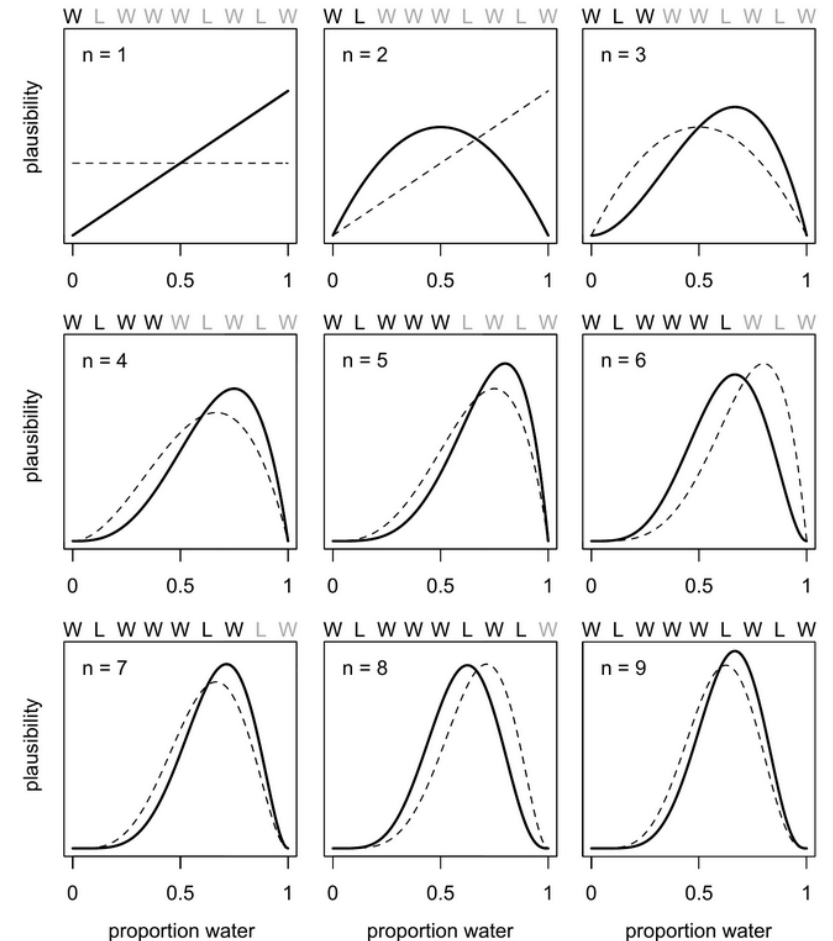
$$\beta_{ai} \sim N(\mu = 0, \sigma = 1) \quad \text{weakly informative prior}$$

$$\beta_m \sim N(\mu = 3, \sigma = 0.5) \quad \text{informative prior}$$

Priors regularize: surprising results require strong evidence.
Additionally, priors can be refined iteratively from study to study.

Updating Priors

Consider data about a globe toss resulting either in water (W) or land (L) to estimate the probability of landing in water when selecting a random place on the globe.



Updating Priors

Calculating $P(d)$ requires integrating over *every possible combination* of coefficient values. For even a couple of predictors, **there is no closed-form algebraic solution** for this in the general case.

- Monte Carlo Markov Chains instead **draws samples** from the distribution even without knowing it.
- You don't need to evaluate $P(d)$ to know the *relative* probability of two parameter combinations. If one combination fits the data and prior better than another, MCMC will “visit” it more often.
- Visualization: <https://chi-feng.github.io/mcmc-demo/app.html>



Bayesian Workflow

The Bayesian Workflow

The complete Bayesian workflow for evaluating a hypothesis comprises the following steps:

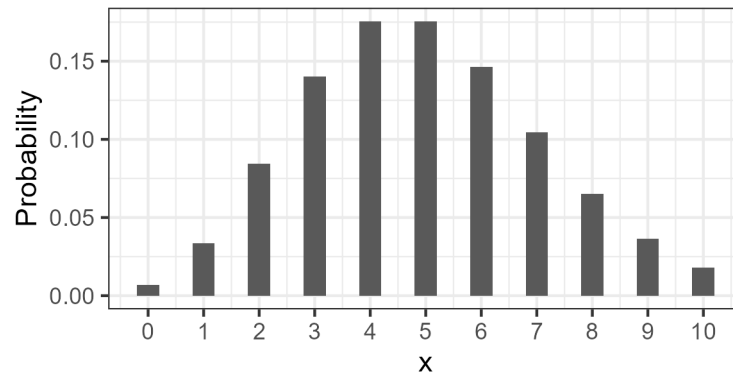


**Selecting a
distribution
family**

that describes
how the data
was created

Determining a Distribution Family

Poisson distribution ($\lambda=5$)



$$\mu = np \quad n \rightarrow \infty$$

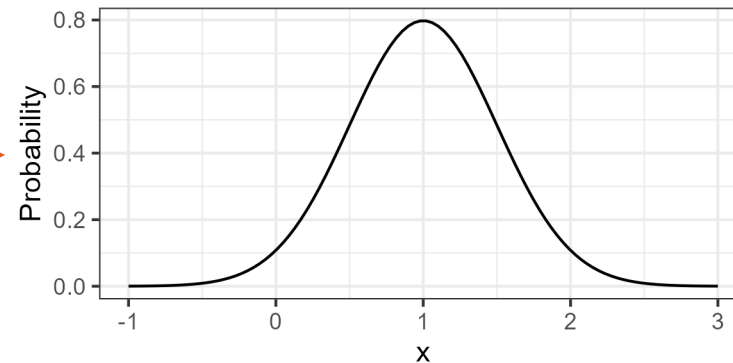
$$\sigma^2 = \mu$$

$$\mu \rightarrow \infty$$

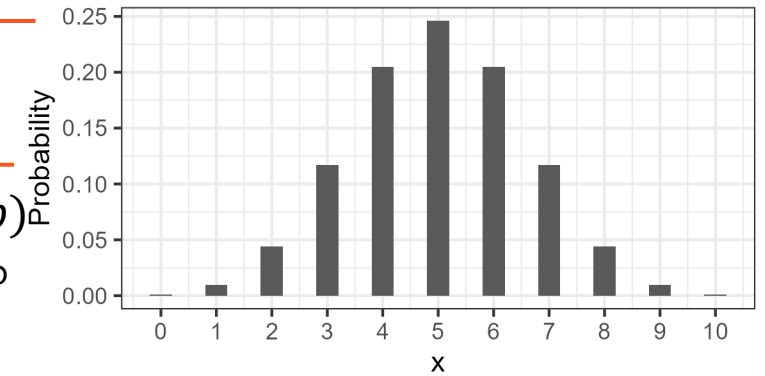
$$\mu = np$$

$$\sigma^2 = np(1-p)$$

$$\text{Normal distribution } (\mu=1, \sigma^2=0.5) \quad n \rightarrow \infty$$

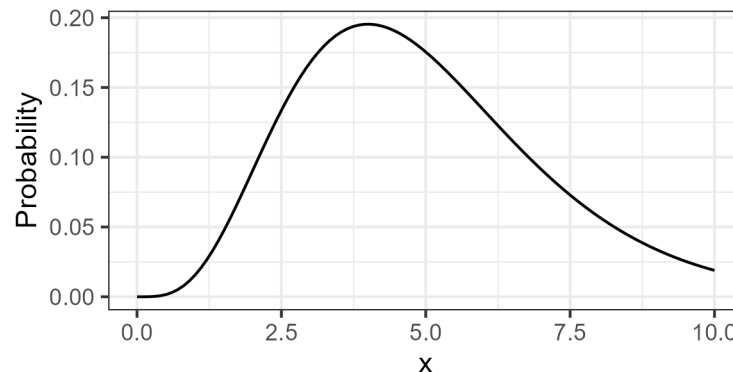


Binomial distribution ($n=10, p=0.5$)

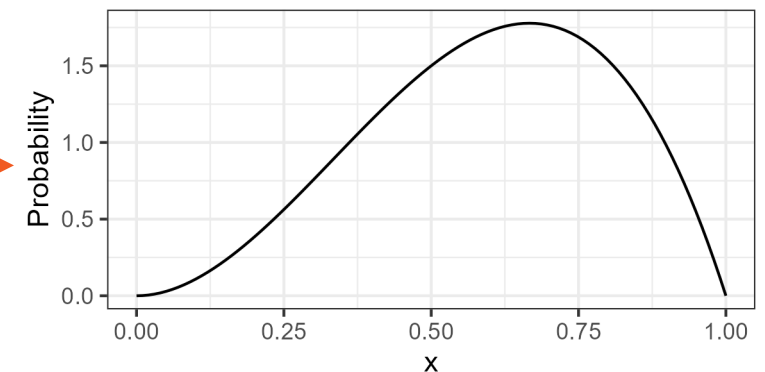


$$\mu = \alpha\beta \quad \sigma^2 = \alpha^2\beta \quad \beta \rightarrow \infty$$

Gamma distribution ($\alpha=5, \beta=1$)



Beta distribution ($\beta=3, \gamma=2$)



$$\frac{X_1}{X_1 + X_2}$$

Determining a Distribution Family

Fully specified, a Bayesian regression model will always start with a likelihood function (i.e., a distribution family) describing the outcome variable.

$$\textit{productivity} \sim \textit{Poisson}(\lambda)$$

$$\log(\lambda) \leftarrow \alpha + \beta_{ai} \cdot \textit{GenAI} + \beta_m \cdot \textit{motivation}$$

$$\alpha \sim N(\mu = 5, \sigma = 2)$$

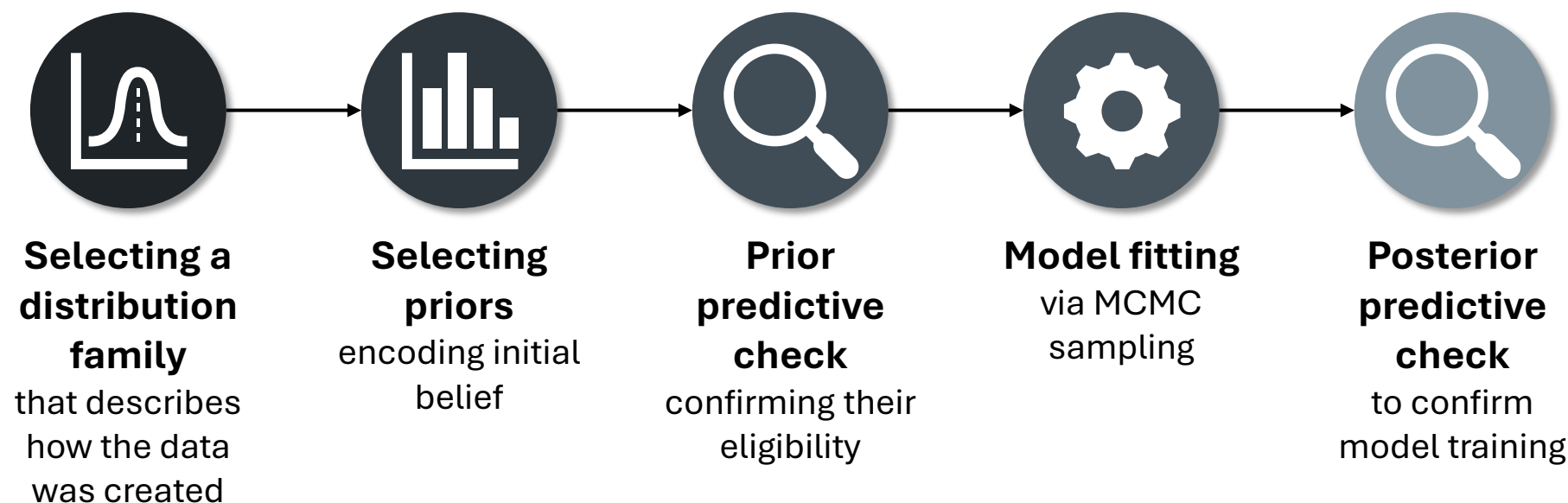
$$\beta_{ai} \sim N(\mu = 0, \sigma = 1)$$

$$\beta_m \sim N(\mu = 3, \sigma = 0.5)$$

This particular model would assume that productivity is Poisson-distributed (e.g., when operationalizing it as the number of commits).

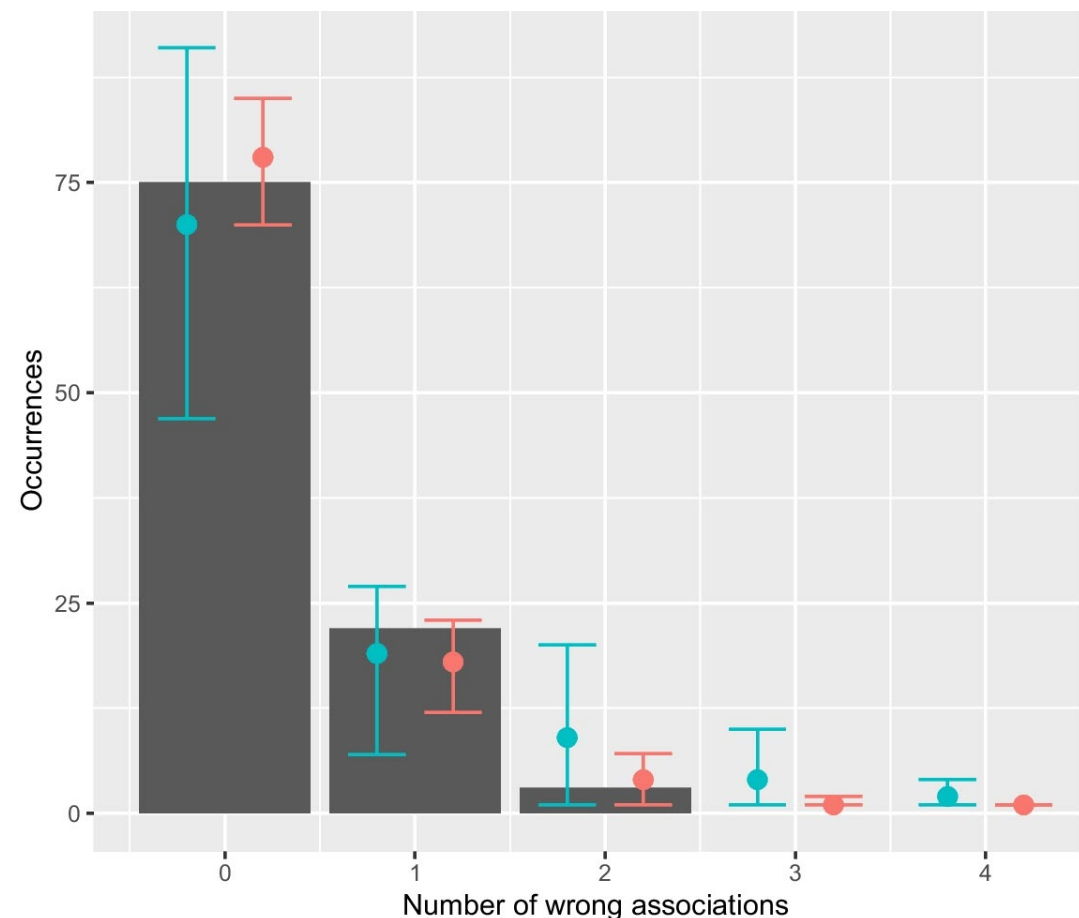
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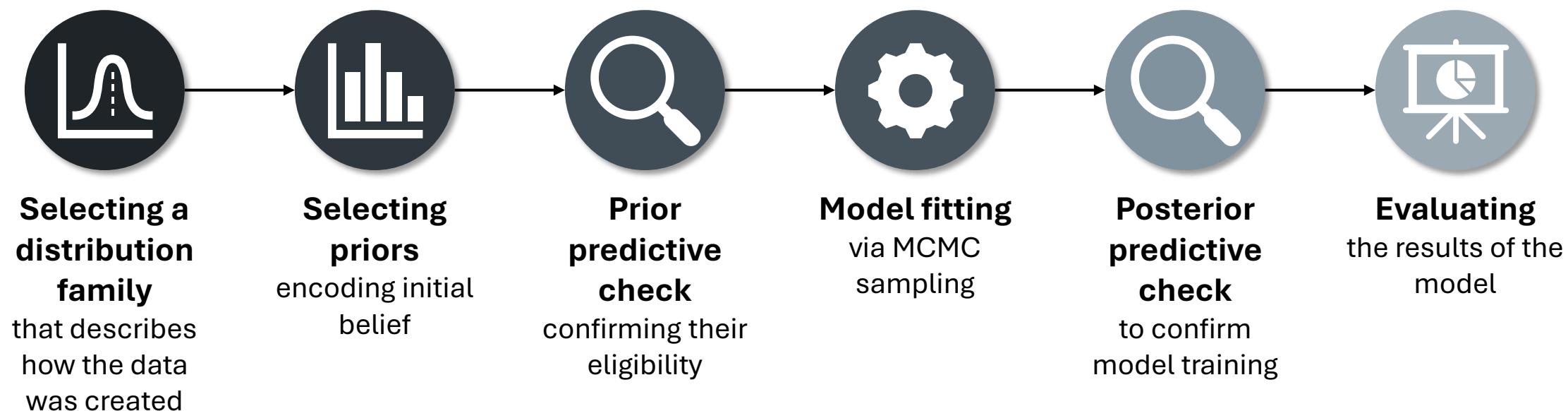
Prior and Posterior Predictive Checks

We can perform a predictive check (i.e., sample from the model) by putting the collected data of the exposure and other covariates in our regression model and checking – given all the uncertainty of the regression coefficients – what the outcome would be.



The Bayesian Workflow

The complete Bayesian workflow for evaluating a hypothesis comprises the following steps:



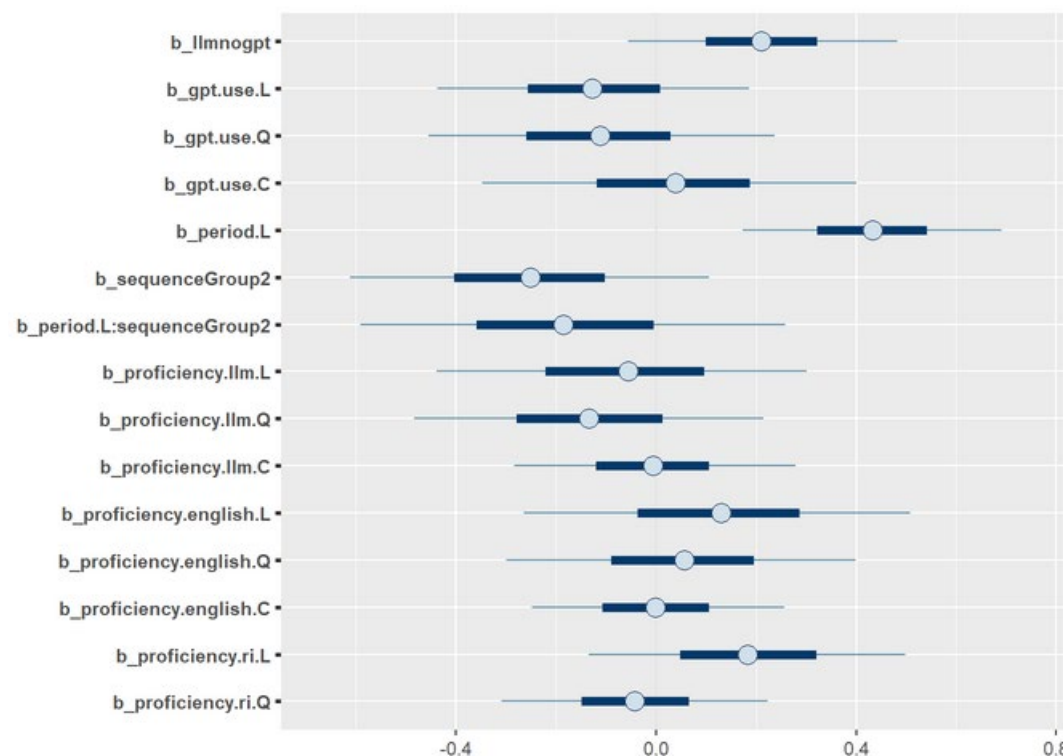
Evaluating Bayesian Output

The result of a Bayesian regression model are **credible intervals** for each regression coefficient representing their *effect on the outcome*. They are set at a certain level (e.g., 95% or 89%) but should not be confused with **confidence intervals**:

- (Bayesian) **credible intervals** mean given the observed data, there is a X% probability that the true parameter value lies within this interval.
- (frequentist) **confidence intervals** mean when repeating this experiment infinitely many times and computing a confidence interval each time, X% of those intervals would contain the true parameter value.

Evaluating Bayesian Output

Credible intervals of regression coefficients indicate, which variable affects the outcome significantly.



Here, significance is achieved if the interval is consistently above or below 0.

Evaluating Bayesian Output

If the distribution family of the regression model requires a link function, the credible intervals of regression coefficients cannot be interpreted directly anymore.

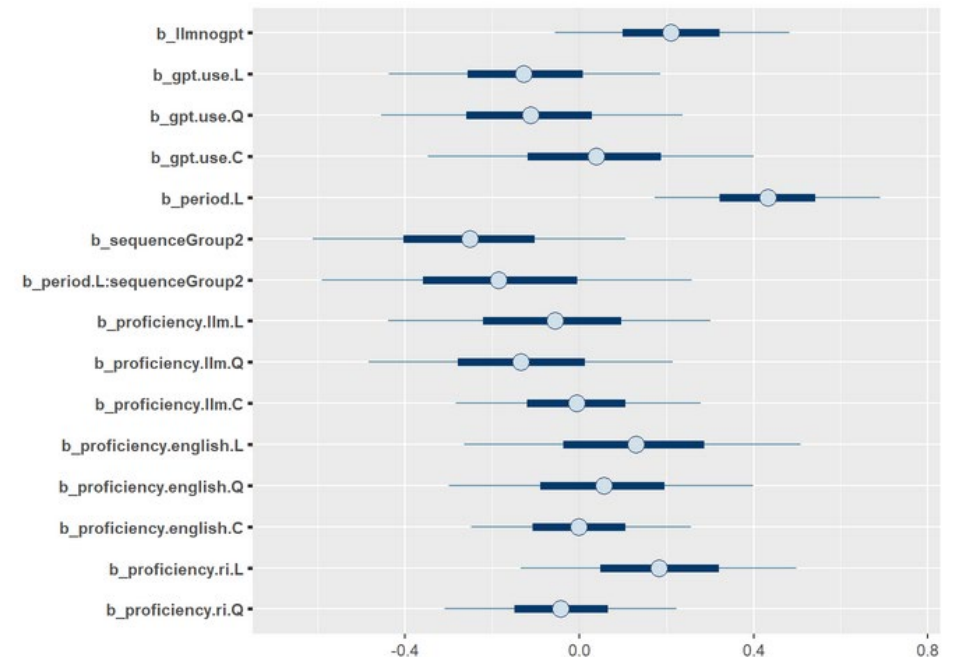
productivity $\sim \text{Poisson}(\lambda)$

$\log(\lambda) \leftarrow \alpha + \beta_{ai} \cdot \text{GenAI} + \beta_m \cdot \text{motivation}$

$\alpha \sim N(\mu = 5, \sigma = 2)$

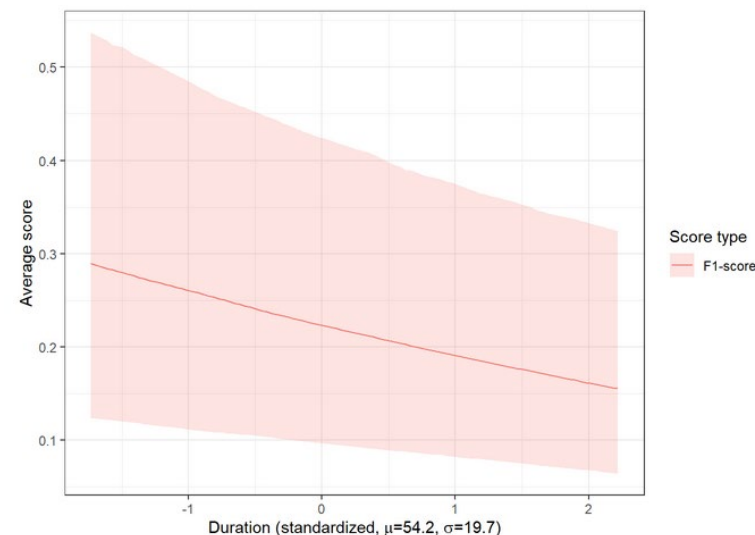
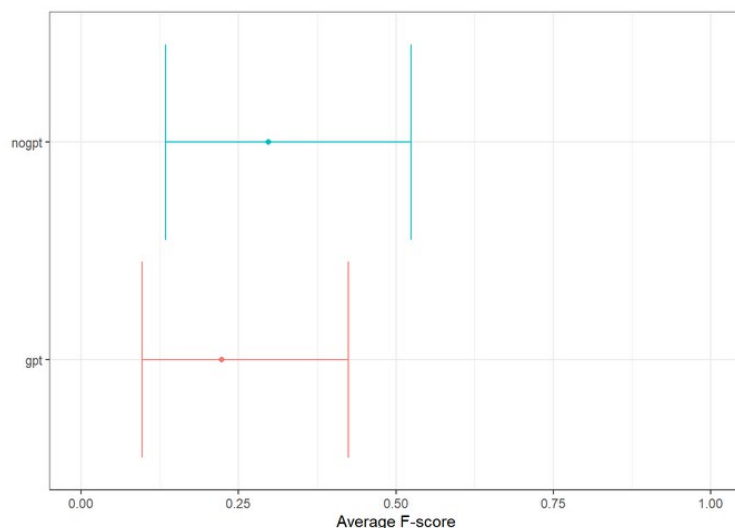
$\beta_{ai} \sim N(\mu = 0, \sigma = 1)$

$\beta_m \sim N(\mu = 3, \sigma = 0.5)$



Evaluating Bayesian Output

The most natural result that a Bayesian regression model produces is a **marginal effect**, i.e., the isolated effect of one variable while keeping the others constant at a representative level. It encodes all uncertainty by all variables and serves to illustrate the true effect.





Limitations

Subjectivity of Priors

Prior probability distributions are – by nature – **subjective** and there is no standard procedure to ensure their “correctness”. However, they are

- **explicit**, i.e., clearly communicated, and
- **changeable**, i.e., possible to update given new information.

In general, a sufficient number of samples will “overwhelm” even strict priors. On the positive side, they serve as a bridge to consider qualitative domain knowledge.

Limitations

Despite all its advantages, Bayesian data analysis has several drawbacks, including:



Slow to compute

Converging a model requires significantly more time than a NHST



Hard to learn

Bayesian data analysis does not have out-of-the-box methods



Hard to explain

Bayesian data analysis is not yet well-established in software engineering



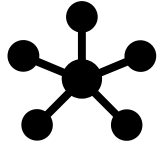
Requires more communication

There are more reportable details to Bayesian data analysis

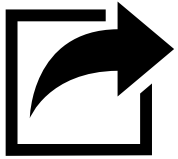


Conclusion

How to Good Science



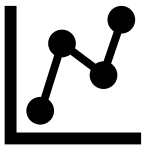
Internal validity requires visualizing and communicating your causal assumptions, and closing any backdoor paths by controlling confounders



External validity requires awareness of potential moderators that also affect trial eligibility



Construct validity requires operationalizing constructs with an encompassing and representative set of indicators



Conclusion validity requires analyzing the data true to its nature